AITEXM ROBOT CMTPD 100W 5A USB-C Fast Charge Trigger Board Module

CMTPD 100W 5A USB-C Fast Charge Trigger Board Module User Manual

Brand: AITEXM ROBOT | **Model:** CMTPD 100W 5A USB-C Fast Charge Trigger Board Module

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Introduction

This manual provides detailed instructions for the AITEXM ROBOT CMTPD 100W 5A USB-C Fast Charge Trigger Board Module. This versatile module is designed to trigger specific voltages from USB Power Delivery (PD) and Quick Charge (QC) compatible power sources, enabling flexible power solutions for various applications. Please read this manual thoroughly before use to ensure proper operation and to prevent damage to the module or connected devices.

Product Features

- Integrated TYPE-C interface for robust and durable connection.
- High-durability USB 5A large current interface.
- Supports external fast switch buttons and software security lock.

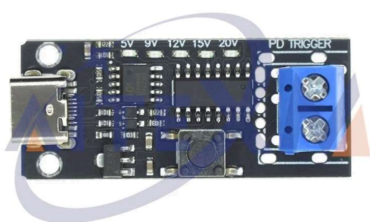
- Features voltage detection and automatic polling.
- High withstand voltage (above 30V) for aging tests.
- Adjustable voltage jump time and low voltage flashing prompt.
- Supports PD3.0, PD2.0, QC, FCP, AFC voltage decoy output with free mode switching.
- Key lock function to prevent accidental voltage changes.
- LED indicators for 5V/9V/12V/20V fast setting and output status.
- Saves the last deception setting, restoring it upon power-on.
- Ultra-low power consumption with high current support, up to 100W output.
- Supports various automatic polling modes: to highest voltage and hold, to highest then 5V, or infinite cycle test.

Specifications

Attribute	Value
Brand Name	AITEXM ROBOT
Model	CMTPD 100W 5A USB-C Fast Charge Trigger Board Module
Type	Module
Application	Computer
Condition	New
Origin	Mainland China
Package	SMD
High-concerned chemical	None
is_customized	Yes
Supported Protocols	PD3.0, PD2.0, QC, FCP, AFC
Output Voltages	5V, 9V, 12V, 15V, 20V (specific range depends on power supply)
Max Power Output	100W (with E-Marker cable/source)
Max Current Output	5A (with E-Marker cable/source)
Withstand Voltage	Above 30V
Default Polling/Jump Time	0.5 seconds (adjustable)

Module Variants and Components

The CMTPD module is available in several variants, primarily differing in their output interfaces. All variants share the core functionality of PD/QC triggering.



Terminal Output Variant: Features screw terminals for direct wire connections.

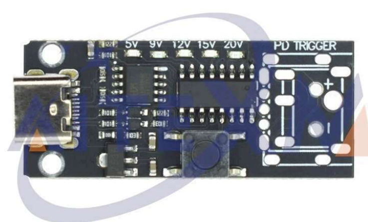
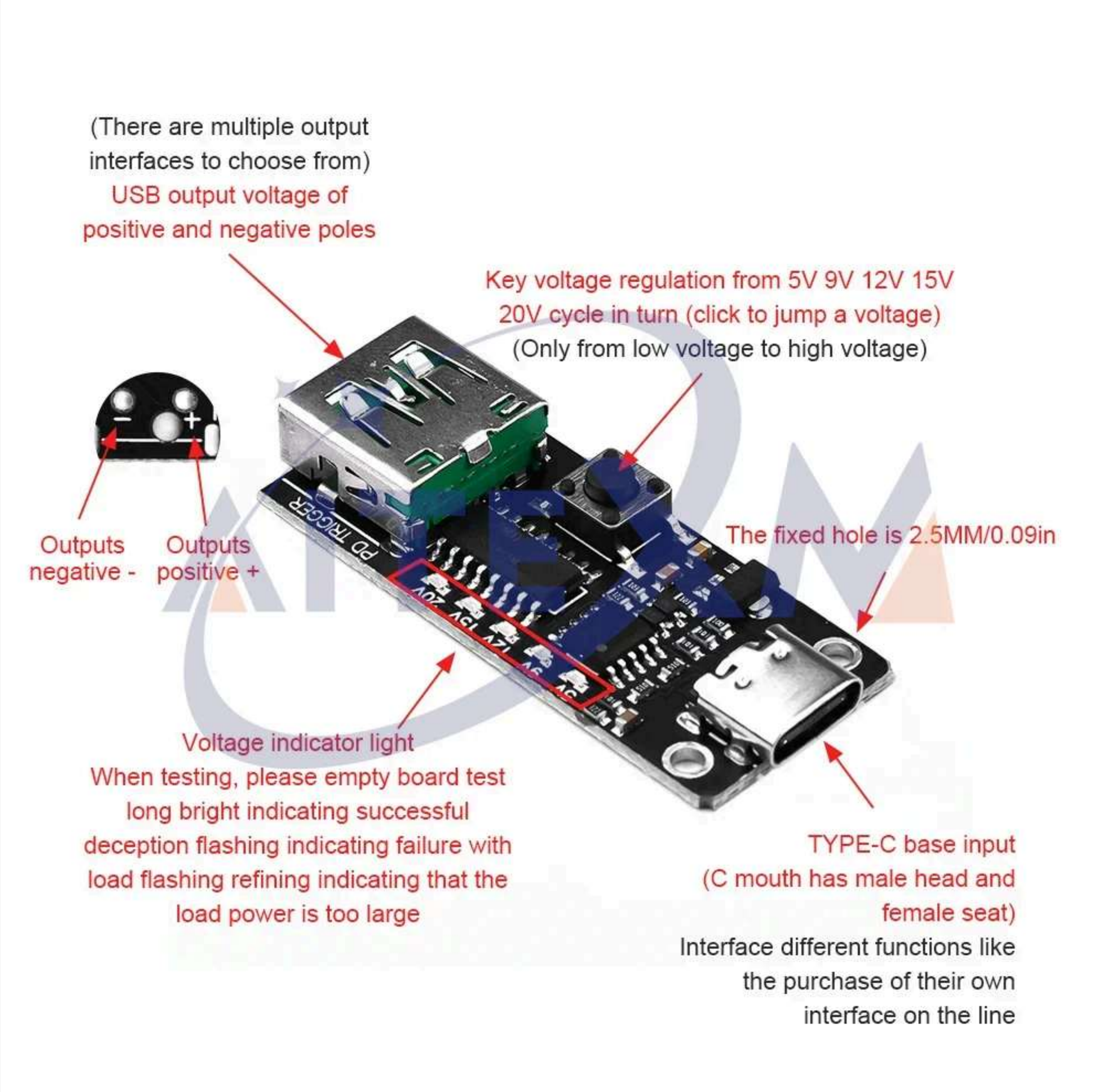


Plate Output Variant: Offers solder pads for custom output connections.

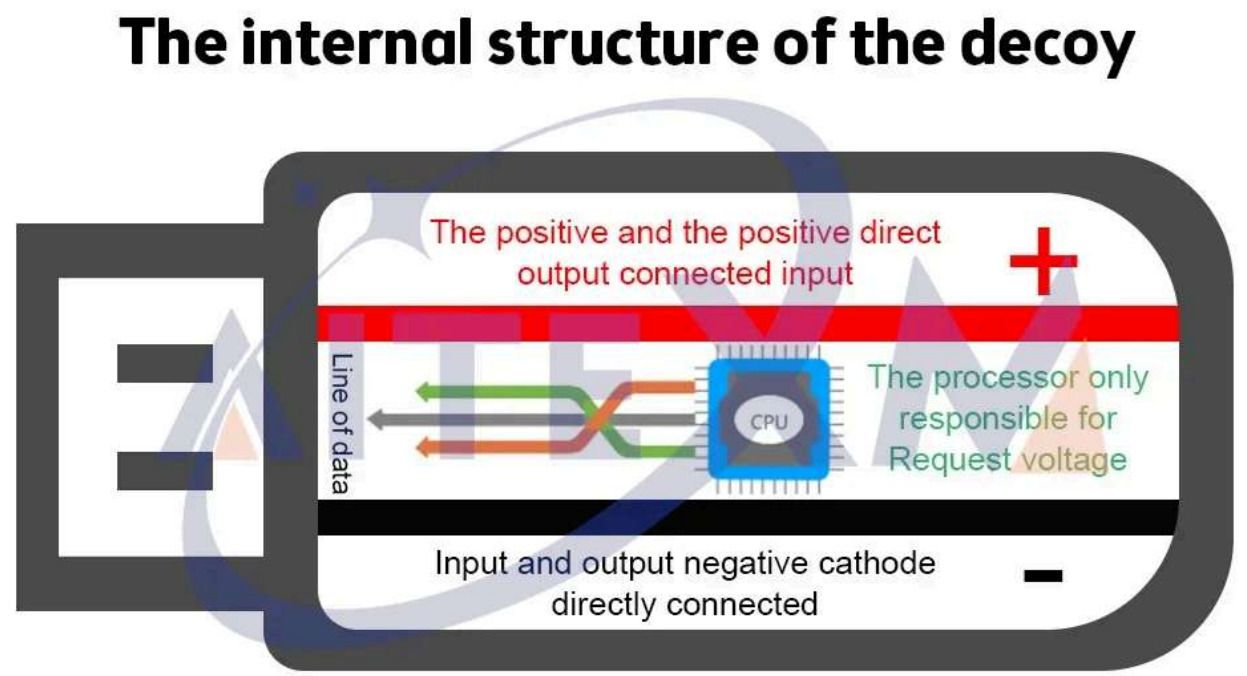
CMTPD module with USB-A Output

USB-A Output Variant: Provides a standard USB-A port for output.

Component Overview



Component Overview Diagram: Illustrates key components including the PD/QC protocol chip, logical control memory chip, 5A current Type-C input, high voltage power supply, voltage adjustment key, and output terminals.



USB-A Variant Feature Diagram: Highlights outputs, voltage regulation key, voltage indicator light, Type-C base input, and fixed mounting hole on the USB-A output variant.

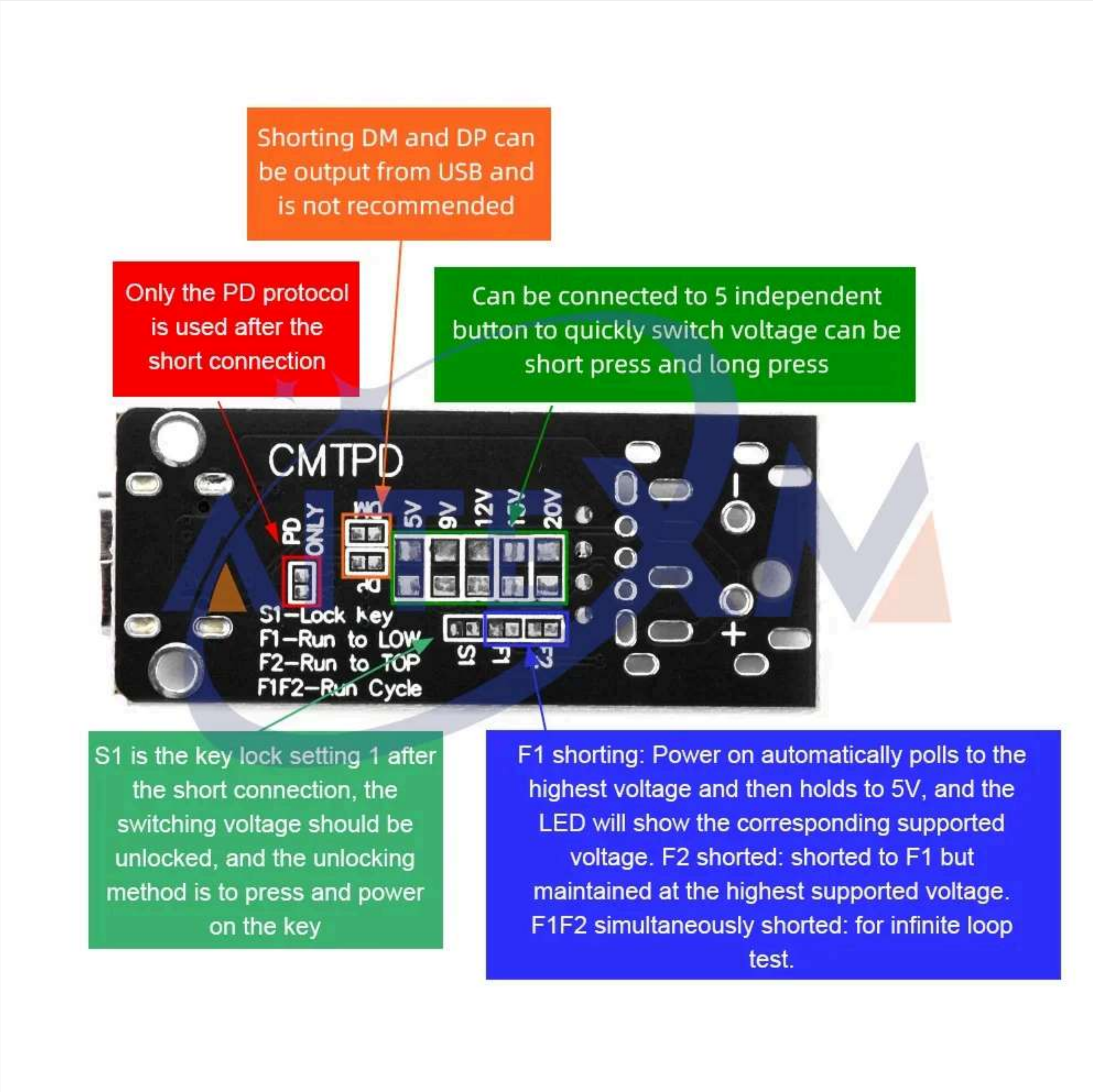
Setup Instructions

1. **Power Source Requirement:** The CMTPD module requires a charging head or power bank that supports QC (Quick Charge) or PD (Power Delivery) fast charging protocols. Verify your power source's specifications to ensure it provides the desired voltage outputs (e.g., 5V, 9V, 12V, 15V, 20V).
2. **E-Marker Cable Consideration:**
   - For power outputs exceeding 60W (up to 100W), an E-Marker enabled USB-C cable is required.

- If using the Type-C female input to trigger 5A current, the connecting cable must support E-Marker.
- The Type-C male input module variants come with an analog E-Marker built-in.

3. **Connecting the Power Source:** Connect your QC/PD compatible power source to the Type-C input port on the CMTPD module.
4. **Connecting the Load:** Connect your target device or circuit to the output terminals of the CMTPD module. Ensure correct polarity (+ and -) for terminal and plate output variants. For USB-A output variants, simply plug in your USB-A device.
5. **Initial Voltage Check:** Before connecting sensitive equipment, it is recommended to perform an empty board test to confirm the output voltage. The voltage indicator lights will show the current output.

Control and Configuration Diagram



Control and Configuration Diagram: Shows the location of shorting contacts (PD ONLY, DM/DP, S1, F1, F2) and the voltage adjustment key.

Operating Instructions

Voltage Selection

The module supports 5V/9V/12V/15V/20V output. Use the onboard key to cycle through these voltages. Each press of the key will jump to the next voltage in sequence (low to high). The corresponding LED indicator will light up to show the selected voltage.

Advanced Configuration (Short Circuit Contacts)

The module features several short circuit contacts for advanced configuration. These contacts are typically small pads on the PCB that can be bridged with solder or a jumper wire.

- **PD ONLY Short Circuit Contact:** Bridging this contact will configure the module to support only the PD (Power Delivery) protocol, ignoring other fast charging protocols like QC, FCP, and AFC.
- **DM and DP Short Circuit Contacts:** Shorting DM and DP allows data lines to be input from the USB. This is generally not recommended unless specifically required for your application, as it can interfere with standard operation.
- **5 Independent Key Interfaces:** These interfaces allow connection of external buttons to quickly switch voltage. They can be configured for short press, long press, or continuous short circuit. The upper ends of these interfaces are grounded.
- **S1 Short Contact (Key Lock):** Bridging S1 enables a key lock function. After shorting S1, the voltage switching function is locked. To unlock, press and hold the onboard key while powering on the module. This lock needs to be re-applied each time the module is powered on.
- **F1 Short Circuit (Automatic Polling to Highest then 5V):** When F1 is shorted, upon power-on, the module will automatically poll through all supported voltages to find the highest available, then settle and maintain 5V. The LEDs corresponding to all supported voltages will light up; if a voltage is not supported, its LED will flash.
- **F2 Short Circuit (Automatic Polling to Highest and Hold):** When F2 is shorted (in conjunction with F1), the module will poll to the highest supported voltage and maintain that highest voltage output.
- **F1 and F2 Shorted Simultaneously (Infinite Loop Test):** Shorting both F1 and F2 initiates an infinite loop test. The module will continuously cycle through voltages from low to high. This is useful for testing and aging.

Adjusting Polling/Voltage Jump Time (for F1/F2 modes)

The default polling and voltage jump time is 0.5 seconds. To modify this:

1. Press and hold the onboard key **without powering on** the module.
2. The first LED light will flash. If you release the key now, the key lock (S1 function) is unlocked.
3. Continue to press and hold the key until all LEDs flash. Release the key to enter the time setting mode. The switching time will reset to 0.5 seconds.
4. Each subsequent press of the key will add 0.5 seconds to the jump time. The LEDs will display the corresponding time in binary mode (e.g., 1st LED for 0.5s, 2nd for 1s, etc.). You can add up to 30 times (maximum 15.5 seconds).
5. If no key is pressed within 3 seconds, the time setting will automatically exit, and the new setting will be saved. All LEDs will flash briefly to confirm.

MAINTENANCE

- **Keep Dry:** Avoid exposing the module to moisture or liquids, which can cause short circuits and damage.
- **Cleanliness:** Keep the module free from dust and debris. Use a soft, dry brush or compressed air for cleaning. Do not use harsh chemicals.
- **Handle with Care:** This is an electronic component. Avoid dropping or subjecting it to physical shock.
- **Storage:** Store the module in a cool, dry place when not in use.

Troubleshooting

- **No Output Voltage / Incorrect Voltage:**
  - Ensure your power source (charger/power bank) supports QC or PD fast charging and can provide the desired voltage.
  - Verify that the Type-C input cable is E-Marker enabled if you require more than 60W or 5A output.
  - Check the selected voltage using the onboard key and LED indicators.
  - If the key lock (S1) is enabled, ensure you unlock it by pressing the key during power-on.
- **Voltage Indicator LED Flashing Rapidly:**
  - If the LED flashes rapidly during a load test, it indicates that the connected load's power requirement is too large for the current power source or the module's capabilities. Reduce the load or use a more powerful/compatible power source.
- **Module Not Responding:**
  - Disconnect and reconnect the power source.
  - Ensure there are no unintended short circuits on the configuration pads (PD ONLY, DM/DP, S1, F1, F2).

User Tips

- Always confirm the output voltage with a multimeter before connecting sensitive devices, especially after changing settings or power sources.
- When experimenting with different voltages, start with the lowest supported voltage and gradually increase to avoid damaging your load.
- For permanent installations, consider using the S1 key lock feature to prevent accidental voltage changes.
- If using external buttons for voltage switching, ensure they are properly wired to the designated independent key interfaces.

Warranty and Support

For warranty information, technical support, or service inquiries, please contact the seller or manufacturer directly. Keep your purchase receipt or order details handy as proof of purchase.

Related Documents - CMTPD 100W 5A USB-C Fast Charge Trigger Board Module

	CMTPD 100W 5A Type-C PD/QC Fast Charge Trigger Board Module Specifications and operational guide for the CMTPD 100W 5A Type-C Fast Charge Trigger Board Module, featuring PD 3.0, PD 2.0, QC, FCP, AFC support for voltage decoy and testing.
	Diymore LX-PD01-V6.2 PD/QC Decoy Trigger Board - Fast Charging Module User Guide Detailed technical specifications and usage instructions for the Diymore LX-PD01-V6.2 PD/QC Decoy Trigger Board, supporting PD3.0/PD2.0 and QC3.0/QC2.0 fast charging protocols for USB Type-C devices. Learn how to set voltages from 5V to 20V.
	Baseus Pudding Series Fast Charging Cable Type-C to Type-C 100W მომხმარებლის სახელმძღვანელო ეს დოკუმენტი შეიცავს ინსტრუქციებს და უსაფრთხოების მითითებებს Baseus Pudding Series Fast Charging Cable-ისთვის, 100W Type-C-დან Type-C-მდე კაბელისთვის, რომელიც მხარს უჭერს მონაცემთა გადაცემას და აღჭურვილია E-Marker ჩიპით უსაფრთხო კვებისათვის.
	elough Charger User Manual: Usage Instructions and Important Notes Comprehensive user manual for elough chargers, detailing usage instructions, plug selection for EU, US, UK, and essential safety notes for optimal and safe charging.